

Chemotherapy of Antiprotozoal Agents

Amoebiasis | Giardiasis | Trichomoniasis | Trypanosomiasis | Leishmaniasis | Toxoplasmosis

Leslie Goldstein, Pharm.D.

Associate Professor of Pharmacology

Department of Clinical Specialties

Rockefeller Building, Room 204

lgolds01@nyit.edu



Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Following completion of the preparation material, students should be able to:

1. List the clinically relevant pharmacokinetic properties of the various chemotherapeutic agents used in the treatment of protozoan infections and give examples.
2. Apply the structure activity relationships and mechanisms of action of the various chemotherapeutics to their use in the treatment of specific protozoan infections and give examples.
3. Relate the mechanisms of adverse effects and drug interactions to the patients' drug therapy.
4. Write a second order, board-style question to assess your understanding of the antiprozoal agents. (Please see the Practice Questions PDF for examples.)

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Optional materials:

- Video Lecture | Dr. Goldstein's Word handout | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links

ScholarRx

Name of Discipline/ Organ System: General Microbiology > Infectious Agents > Parasites and Fungi

Title of Brick(s): Protozoa and Ectoparasites

Link: <https://exchange.scholarrx.com/brick/protozoa-and-ectoparasites>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 52: Antiprotozoal Drugs > Malaria

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382§ionid=281755910#1204144707>

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology 8e, 2023; Chapter 35 Antiprotozoal Drugs > Malaria <https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329448&bookid=3222>

Access Medicine Katzung & Trevor's Pharmacology: Examination & Board Review, 13e, 2021; Chapter 52: Antiprotozoal Drugs > Malaria

<https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3058§ionid=255307694>

Key points: Pathogenic protozoa

- A wide variety of protozoal parasites can be transmitted to human hosts by insect vectors, directly from other mammalian reservoirs, or from one person to another. The immune system is critical in protecting against the pathological effects of infection.
- Amoebiasis caused by *Entamoeba histolytica*, spread by the fecal-oral route, infects the intestines and can lead to amoebic dysentery or liver abscess.
- *Giardia lamblia* is transmitted through contaminated water and leads to diarrhea, cramping, and malabsorption.
- *Trichomonas vaginalis* is the most common nonviral sexually transmitted infection worldwide. Infections range from an acute severe inflammatory disease to asymptomatic carrier state.
- African trypanosomiasis (sleeping sickness) is caused by *Trypanosoma brucei* spread by tsetse flies and can lead to CNS infection and coma. Chagas disease, caused by *Trypanosoma cruzi* and transmitted by triatomine bugs, can lead to severe heart and digestive disease.
- *Leishmania* species transmitted by sandflies cause cutaneous / mucocutaneous and visceral disease. Visceral leishmaniasis is life-threatening.
- *Toxoplasma gondii* is an obligate intracellular parasite transmitted to humans from cats (the only definitive host), but more commonly via undercooked infected meat products. The infection is often asymptomatic but can cause serious complications in pregnant individuals and immunocompromised patients.
- Drug therapy treats infection and reduces transmission. Effective vaccines against protozoal infections are not available.

Key points (Starred drugs are in bold font.)

Antiprotozoal drugs target protozoal enzymes, metabolic processes, and organelles not present in humans and, therefore, are more selective for parasitic cells than for human cells. In addition, many protozoa need anaerobic or microaerophilic environments to function, whereas human cells require oxygen.

- The drugs interfere with protozoal nucleic acid synthesis, inhibit protein synthesis or metabolic processes, disrupt protozoal organelles and/or membrane integrity.
- **Metronidazole**, tinidazole, secnidazole, and nitazoxanole are nitroimidazole compounds active against anaerobic and microaerophilic parasites. They are converted by parasitic pyruvate:ferrodoxin oxidoreductase to toxic nitro free radical intermediates, which cause death of the parasite. Resistance, though uncommon, arises through decreased activation or increased inactivation of drug.
- **Metronidazole PK**: several routes of administration, widely distributed, hepatic metabolism, bioavailability nearly 100%, and eliminated in urine/feces as active drug and metabolites.
- **Metronidazole** is used in infants, children, and adults for the treatment of amoebiasis, giardiasis, trichomoniasis, and anaerobic bacterial infections.
- Tinidazole is used orally or intravaginally for adults and children ≥ 3 years old for the same parasitic infections.
- Secnidazole is a long-acting nitroimidazole labeled for oral single-dose treatment of bacterial vaginosis and vaginal trichomoniasis in patients ≥ 12 years old.
- Nitazoxanide is a broad-spectrum antiparasitic agent used orally for the treatment of and cryptosporidiosis-associated diarrhea, giardiasis, and other parasitic infections.
- **Paromomycin** is an aminoglycoside (protein synthesis inhibitor), which is not absorbed from the GI tract. It is administered orally for intestinal amebiasis and certain other luminal parasitic infections.

Key points: Trypanosomiasis

- Sleeping sickness, caused by *Trypanasoma brucei*, is endemic in some regions of sub-Saharan Africa. *T. brucei gambiense* causes human West African trypanosomiasis – gambiense HAT. It has an incubation period of weeks to months and is usually slowly progressive. *T. brucei rhodesiense* causes human East African trypanosomiasis – rhodesiense HAT. It has an incubation period of less than three weeks and is usually rapidly progressive within weeks. Gambiense HAT accounts for a larger burden of disease than rhodesiense HAT.
- The approach to treatment selection is guided by patient age and weight, drug availability, neurological signs and symptoms, and CSF findings.
- **Fexinidazole** is an oral drug preferred for both gambiense and rhodesiense early-stage (hemolymphatic) HAT in children ≥ 6 years old, ≥ 20 kg. It is oral, safer than the other drugs, and crosses the blood-brain barrier.
- Children < 6 years old, < 20 kg with early-stage gambiense HAT are treated with IM or IV **pentamidine** and, for rhodesiense HAT, with IV **suramin**.
- Late-stage (meningoencephalitic) gambiense HAT, nonsevere disease, is oral **fexinidazole**. Oral **nifurtimox** with IV **eflornithine** combination therapy (NECT) is more effective in severe disease and the preferred treatment.
- Late-stage rhodesiense HAT is treated with the arsenical **melarsoprol**, which is very toxic.

Key points

- **Fexinidazole**, like other nitroimidazoles, is activated by parasite nitroreductase to nitro free radicals that damage DNA and proteins (the proposed mechanism). PK: oral with the main meal of the day, widely distributed, extensively metabolized via CYPs to active metabolites, half-life between 9 and 27 hours. Adverse effects include neuropsychiatric effects, nausea/vomiting, headache, dizziness, tremor, neutropenia, and QT interval prolongation. Follow-up clinical evaluation should be performed at 6, 12, 18, and 24 months after completion of therapy or at any time symptoms recur due to risk of recurrence, which has been observed in patients with severe late-stage gambiense HAT.
- **Pentamidine** isethionate impairs the parasite's mitochondrial activity and interferes with polyamine synthesis, RNA polymerase, and ribosomal functions and prevents synthesis of proteins, nucleic acids, and phospholipids. PK: slow IV infusion; terminal elimination weeks to months. Common adverse effects include nausea, fatigue, hypoglycemia, and injection site pain and local abscess. Potentially serious toxicities include reversible bone marrow suppression, hepatotoxicity, nephrotoxicity, and neuropathy.
- **Nifurtimox** is an oral trypanocidal imidazole compound taken with food for improved absorption, metabolized in the liver, excreted in urine as metabolites, and has a half-life of ~3 hours. Common adverse effects include nausea/vomiting, headache, dizziness, and polyneuropathy.
- **Eflornithine** is an irreversible inhibitor of ornithine decarboxylase, which rapidly depletes the parasites of polyamines. It is given by slow IV infusion and has a half-life of ~3 hours. Toxicities include fever, pruritis, hypertension, nausea/vomiting/diarrhea, headache, dizziness, seizure, and bone marrow suppression.

Key points

- **Suramin** is the preferred agent for early-stage rhodesiense HAT in children <6 years old, <20 kg. It forms stable complexes with proteins, interacting with a wide variety of biomolecules such as serine oligopeptidase, dihydrofolate reductase, thymidylate synthase, and several glycolytic enzymes, inhibiting energy metabolism in the parasite. It is given by slow IV infusion weekly over four weeks, does not penetrate the CSF, and has a very long half-life (is detectable in urine for 3 months). Early adverse effects include nausea, vomiting, and fatigue. Severe toxicities that occur over time include bone marrow toxicity, nephrotoxicity, peripheral neuropathy, chronic diarrhea, and more. Suramin is also active against the nematode (worm) *Onchocerca volvulus* that causes river blindness. Coinfected patients experience severe immunologic reactions. Therefore, coinfection should be excluded before administering suramin.
- **Melarsoprol** is the only agent effective against late-stage rhodesiense HAT. It is an arsenical prodrug converted to the active and extremely toxic melarsen oxide, which reacts avidly and reversibly with sulfhydryl groups. It forms a melarsen oxide-trypanothione complex that inhibits trypanothione reductase activity. Melarsen oxide inactivates most enzymes of both parasite and host. Melarsoprol reactive encephalopathy occurs in up to 18% of patients and is fatal in many of the cases. The drug is toxic to heart, liver, kidneys, and nerves among others.

Key points

- American trypanosomiasis – Chagas disease – is caused by transmission by the triatomine bug (*Reduviidae* family). It is treated with the oral imidazole compound **benznidazole** (preferred, more tolerable, twice daily dose) or **nifurtimox** (alternative, three times daily dose). They are reduced by parasitic nitroreductase to nitro free radicals that form covalent attachment to biomolecules. Both drugs are trypanocidal against trypomastigote and amastigote forms of *T. cruzi*. Treatment of acute disease reduces parasitemia, morbidity, and mortality. Cure rate is >80%, but with geographic variations. For chronic disease, cures are possible but the drugs are less effective than for acute disease. Adverse effects include nausea/vomiting, headache, dizziness, and polyneuropathy.
- *Leishmania* is a trypanosomatid protozoa transmitted by the bite of certain species of sand fly. Diseases include cutaneous leishmaniasis (Old World and New World varieties) and visceral leishmaniasis (kala-azar). *Leishmania* species differ in their sensitivity to chemotherapeutic agents.
- **Sodium stibogluconate** (most countries) and meglumine (French-speaking countries) are pentavalent antimonial agents widely used first-line agents. Pentavalent prodrugs are reduced to trivalent antimonial by parasitic reductase, which inhibits trypanothione reductase, which interferes with trypanothione redox system, inducing rapid efflux of trypanothione and glutathione from the cell and also inhibiting trypanothione reductase. PK: IV administration, drug is sequestered in tissues, terminal half-life of up to 80 hours. The antimonials are fairly well tolerated. Adverse effects include dose-related muscle aches, nausea/vomiting, fatigue, and rash and nonspecific EKG changes, such as T wave suppression and QT interval prolongation. Additional potentially serious toxicities include bone marrow suppression, hepatotoxicity, nephrotoxicity, and pancreatitis.

Key points

- **Amphotericin B** liposomal IV is recommended in certain areas of India and Nepal where there is resistance to the antimonials. Amphotericin B has a high cure rate in the treatment of mucocutaneous and visceral leishmaniasis. Toxicities are infusion reactions (“shake and bake”), nephrotoxicity, anemia, hypersensitivity reactions, and more.
- **Miltefosine** is the first consistently effective and safe oral drug for visceral, mucosal, and cutaneous leishmaniasis, but efficacy varies significantly by species and by geography. Its mechanism is not known. It is also effective against free-living encephalitic amoeba, such as *Naegleria fowleri*. As a phospholipid, it may interact with phospholipids and steroids in parasitic cell membranes, inhibition of cytochrome c oxidase which inhibits mitochondrial function, and apoptosis-like death. Adverse effects are common, generally mild nausea/vomiting/diarrhea, anemia/decreased platelet count, increased serum aminotransferases, increased serum creatinine, headache, dizziness, rash, urticaria, SJS (rare). *Contraindicated* in all trimesters of pregnancy – teratogenic, fetotoxic. Breast feeding should be avoided. Reduced sperm count, mobility, and morphology, decreased ejaculate volume, scrotal pain and tenderness in males.

Key points

- Toxoplasmosis is an acute, self-limiting illness in most people, which does not require treatment. Immunocompetent persons with primary infection are usually asymptomatic. Toxoplasmosis has a worldwide distribution.
- Pregnant patients are treated for acute acquired toxoplasmosis to prevent transmission to the fetus. Treatment at <14 weeks gestation is with **spiramycin**, an oral macrolide that concentrates in the placenta. If >18 weeks gestation and documented fetal infection, then **pyrimethamine** plus **sulfadiazine** plus **leucovorin** is administered.
- HIV/AIDS patients with cerebral toxoplasmosis are treated with oral pyrimethamine plus sulfadiazine plus leucovorin or TMP-SMX IV, followed by suppressive therapy with pyrimethamine + sulfadiazine + leucovorin.
- Toxoplasmosis prophylaxis in HIV/AIDS patients is oral TMP-SMX.

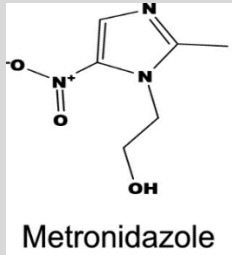
Outline

Drugs for the treatment of:

1. Amoebiasis, giardiasis, trichomoniasis
2. Human African trypanosomiasis (HAT) (sleeping sickness)
 - gambiense HAT
 - rhodesiense HAT
3. American trypanosomiasis (Chagas disease)
4. Leishmaniasis
5. Toxoplasmosis

Some drugs are FDA approved. Others are not FDA approved and are available from the CDC or through an FDA investigational new drug protocol.

Agents used in the treatment of *Entamoeba histolytica*, *Giardia*, and *Trichomonas* infections



a nitroimidazole

*Metronidazole

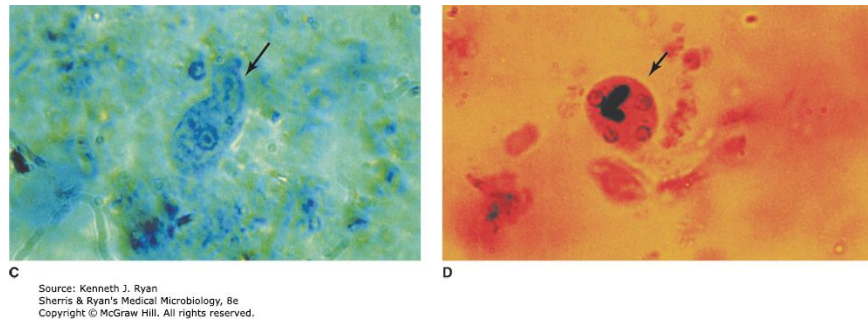
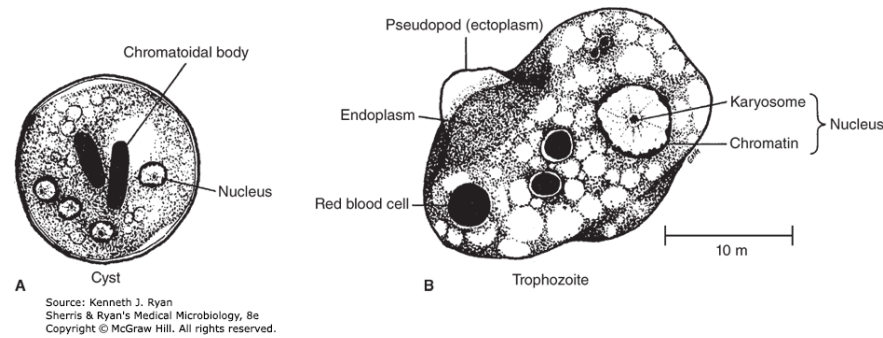
Tinidazole: better tolerated than metronidazole and less frequent dosing

Secnidazole: bacterial vaginosis only

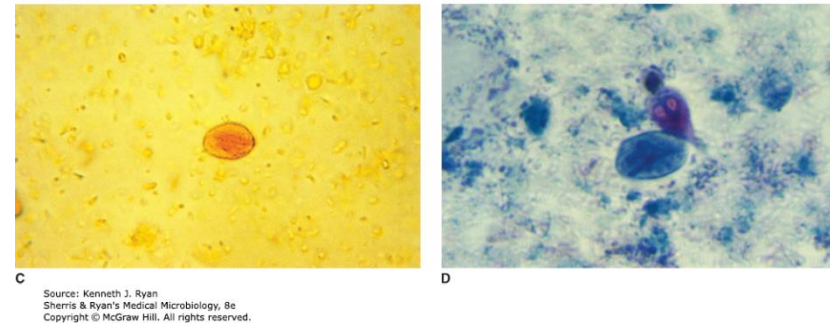
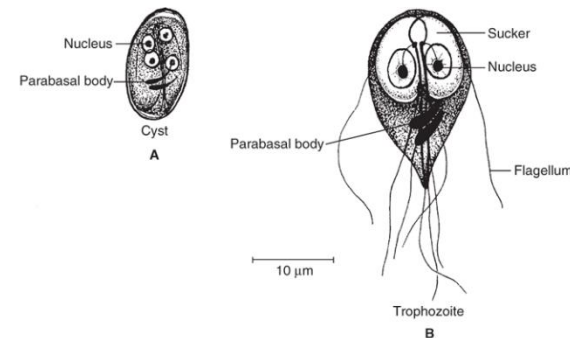
*Paromomycin

Nitazoxanide: Diarrhea caused by giardiasis and cryptosporidiosis (adults, children); and other parasitic diarrheal diseases in children

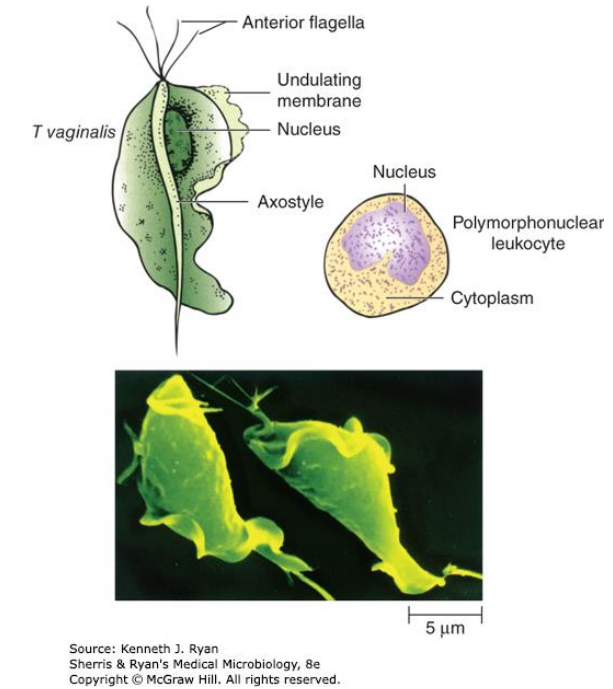
Iodoquinol; Diloxanide furoate: Marketing has been discontinued.



Entamoeba histolytica. A. Cyst structures. B. Trophozoite structures. C. Trophozoite in stool (arrow). D. Cyst (arrow) in stool iodine preparation and cysts in stool iodine preparation.



Giardia lamblia. A. Cyst structures. B. Trophozoite structures. C. Cyst in stool iodine preparation. D. Trophozoite in stool. (CandD, Reproduced with permission from Connor DH, Chandler FW, Schwartz DQ, et al: Pathology of Infectious Diseases. Stamford CT: Appleton & Lange; 1997.)



Trichomonas vaginalis. The parasite and its structures are shown in relation to the size of a polymorphonuclear leukocyte (top). The micrograph below illustrates their use for motility. (Reproduced with permission from Nester EW, Anderson DG, Roberts CE Jr, et al: Microbiology: A Human Perspective, 6th ed. New York, NY: McGraw Hill; 2008.)

Metronidazole Pharmacokinetics

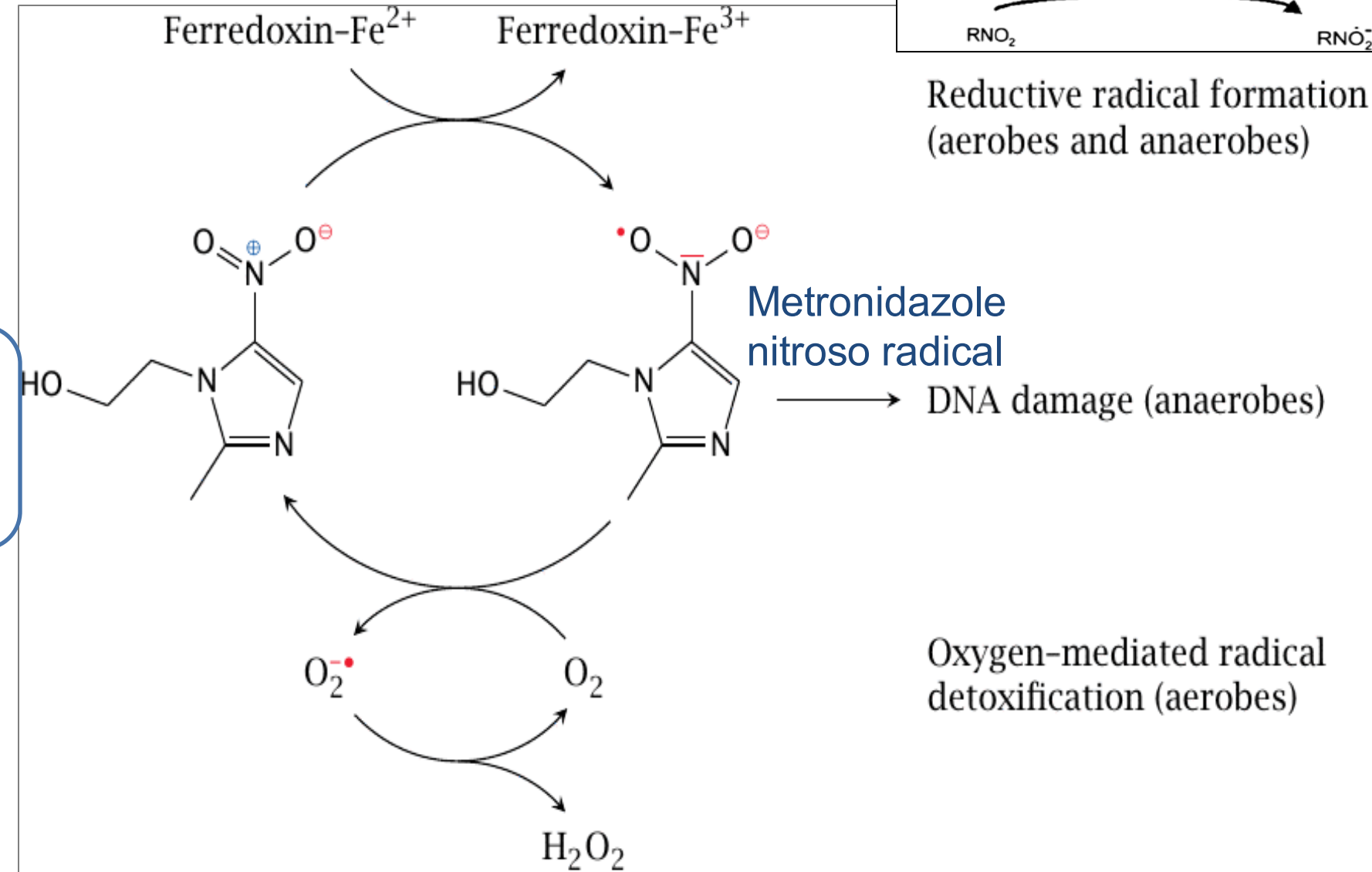
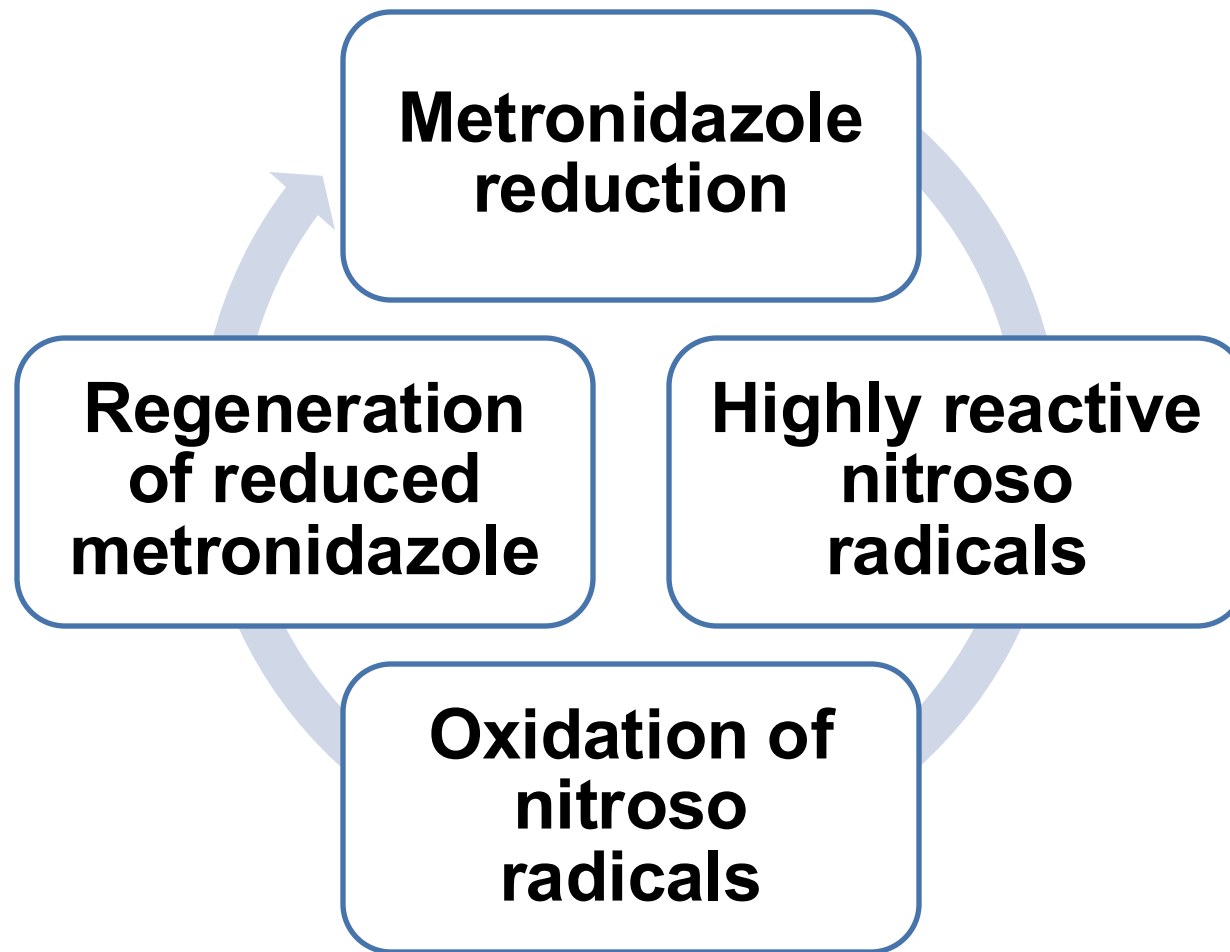
A	Oral, ~100% bioavailability, IV topical, intravaginal → low systemic concentrations	
D	Widely distributed, Penetrates body tissues and fluids well, bile, seminal fluid, bone, liver, and liver abscesses, lung and vaginal secretions, saliva and CSF	
M	Hepatic, CYP2A6, several metabolites including active metabolite	Tinidazole: CYP3A4
E	Urine, some unchanged drug and metabolites	
t_{1/2}	8 h adults; longer in premature and term neonates	Tinidazole: ~13 h
Dosing: Adults: 3 times daily		Tinidazole: once daily

PK relevance:

Metronidazole and tinidazole have good oral bioavailability and penetrate infected tissues in therapeutic concentrations, including abscesses.

Tinidazole has a longer half-life → less frequent dosing.

Metronidazole Regeneration / Resistance Mechanisms



Resistance (uncommon)

<http://watcut.uwaterloo.ca/webnotes/Pharmacology/AntimicrobialChemotherapy.html>

- $\downarrow$ drug activation: Impaired O₂ scavenging capabilities $\rightarrow \uparrow$ local O₂ $\rightarrow \downarrow$ reduction of metronidazole
- Futile cycling: Increased inactivation of the active toxic intermediate
- Lower PFOR levels and ferredoxin levels: These organisms respond at higher drug doses

Metronidazole Clinical Applications

Activity:

Anaerobic and microaerophilic pathogens, such as amitochondriate protozoa *T. vaginalis*, *E. histolytica*, and *G. lamblia*

- Kills *E. histolytica* trophozoites
 - but not cysts
- Acts on organisms in bowel wall and liver
- Eradicates extra-intestinal infection (liver abscess is most common)

Follow up with a luminal amebicide for eradication of parasites.

Therapeutic uses:

Amoebiasis

Trichomoniasis

Giardiasis

Anaerobic *bacterial* infections

Tinidazole equally effective

- amoebiasis, giardiasis, trichomoniasis
- *Helicobacter pylori* (combination Tx)
- Longer $t_{1/2}$; may be better tolerated than metronidazole

Secnidazole

- bacterial vaginosis, trichomoniasis

Metronidazole Side Effects

Generally well tolerated

- Common: headache, nausea, dry mouth, and a metallic taste
- Occasional: vomiting, diarrhea, and abdominal distress, thrush, dark urine
- Dizziness, vertigo
 - Very rarely, encephalopathy, convulsions
- Paresthesia
- **Pregnancy:** Treatment is recommended. Metronidazole crosses placenta but appears to have a low risk to the developing fetus.
- Vaginal administration, tinidazole, and secnidazole should be avoided.

US Boxed Warning

Metronidazole and tinidazole have been shown to be *carcinogenic in mice and rats*.

Practice point:

- Unnecessary use of the drugs should be avoided. Use should be reserved for indicated conditions.

Drug interactions: (not a complete list)

- Metronidazole: Warfarin
- Tinidazole: CYP3A4-mediated interaction

It has been thought that metronidazole ingested with alcohol causes a disulfiram-like reaction.

Reevaluation of available data suggests that this is not the case (Feldman & Jaszczenski, 2023)

Paromomycin: A Luminal Aminoglycoside

PK	Oral; poor absorption – remains in the intestine = luminal; 100% fecal excretion
Mechanism of action	Protein synthesis inhibitor by binding the 16S rRNA of the 30S ribosome interfering with the protein synthesis machinery
Uses	*Acute and chronic intestinal (luminal) amebiasis following nitroimidazole treatment • <i>Cryptosporidium</i>-associated diarrhea in patients with HIV Cryptosporidium is a protozoan parasite spread by the fecal-oral route from contaminated water sources. • <i>Giardia lamblia</i> alternative for mild giardiasis • Hepatic encephalopathy: Reduces systemic ammonia levels by killing ammonia-producing bacteria in the gut. • Topical: cutaneous leishmaniasis IV: visceral leishmaniasis
Side effects	GI effects: abdominal cramps, diarrhea, nausea, vomiting Parenteral: nephrotoxicity, ototoxicity, muscle weakness (as for all aminoglycosides)

Check your knowledge

What are the medical justifications for using two different drugs in the treatment of amoebiasis? What are the drugs described in this PowerPoint?

What are their mechanisms of action?

What other organisms are susceptible to each of the drugs?

Agents for the treatment of Human African Trypanosomiasis (HAT)

Gambiense HAT

T. brucei gambiense

*Fexinidazole

*Pentamidine

*Nifurtimox

*Eflornithine

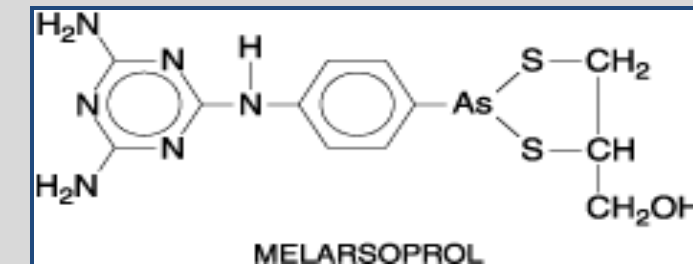
Rhodesiense HAT

T. brucei rhodesiense

*Fexinidazole

*Suramin

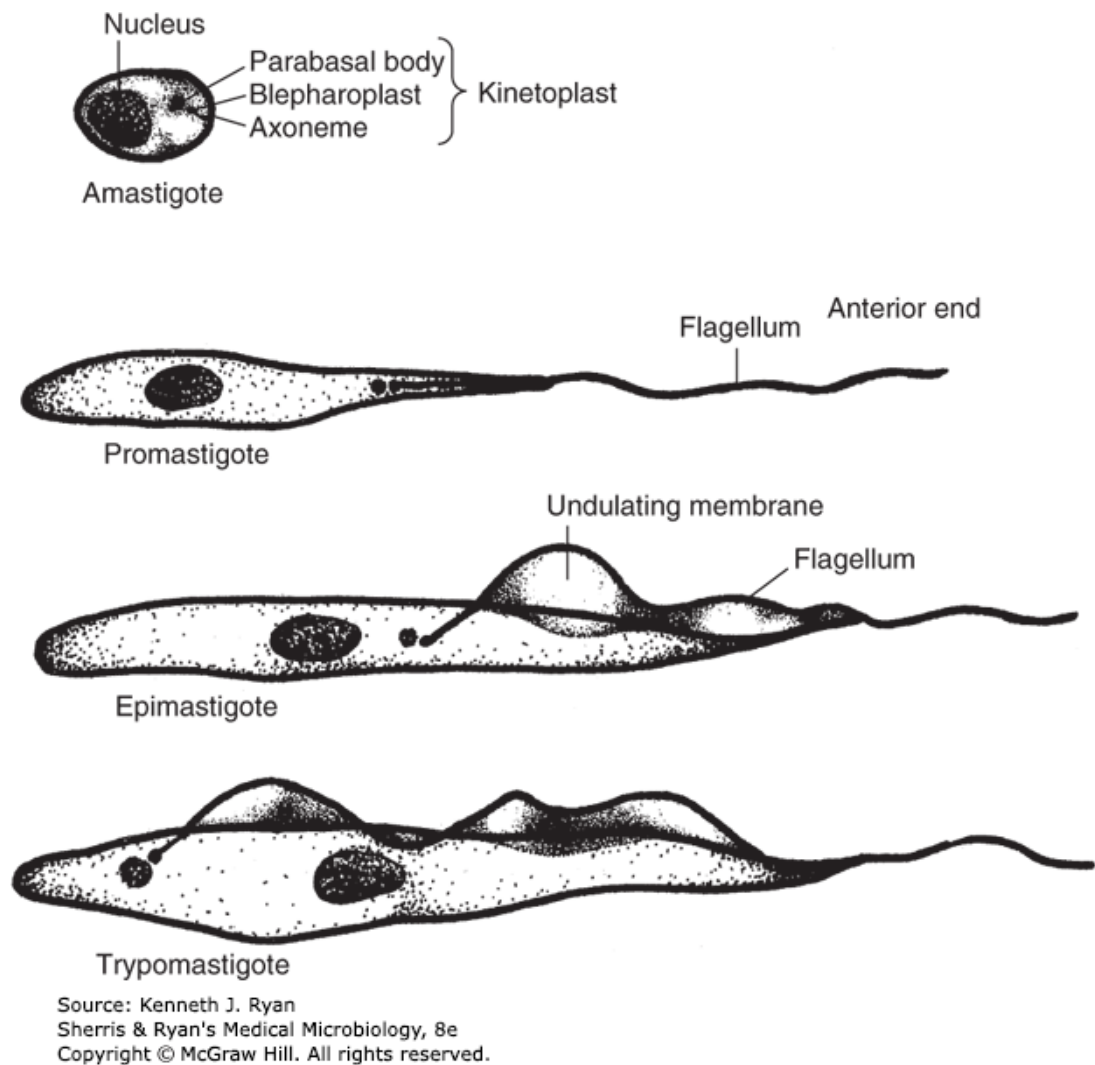
*Melarsoprol



Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period	1 1 H	2 4 He																
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

Arsenic
#33

The approach to treatment selection is based on assessment of neurological signs and symptoms of severe disease. CSF examination is required for patients with signs and symptoms of late-stage severe disease and for patients who cannot be treated with fexinidazole.



Stages in the life cycle of the hemoflagellates (Trypanosomidae).



Source: Kenneth J. Ryan
Sherris & Ryan's Medical Microbiology, 8e
Copyright © McGraw Hill. All rights reserved.

African sleeping sickness. *Trypanosoma brucei* in a routine blood smear. (Reproduced with permission from Nester EW, Anderson DG, Roberts CE Jr, et al: Microbiology: A Human Perspective, 6th ed. New York, NY: McGraw Hill; 2008.)

Trypanosomiasis, caused by the bite of the tsetse fly

Caused ~95% of the cases of human infection in 2022

Gambiense HAT

West African sleeping sickness

- *Trypanosoma brucei gambiense*
Causes the chronic form of the disease.
West and Central Africa
 - Democratic Republic of Congo
 - Angola
 - South Sudan
 - Central African Republic
 - Chad
 - northern Uganda
- 7,000-10,000 cases reported annually

Rhodesiense HAT

East African sleeping sickness

- *Trypanosoma brucei rhodesiense*
Causes acute, rapidly progressive disease
East and Southern Africa
 - Tanzania
 - southern Uganda
 - Malawi
 - Zambia (formerly Northern Rhodesia)
 - Zimbabwe (formerly Southern Rhodesia)
- A few hundred cases reported annually to the WHO.

Caused ~5% of the cases of human infection in 2022

	Gambiense HAT		Rhodesiense HAT
	<6 years old, <20 kg	≥6 years, ≥20 kg	Patients of all ages
Early-stage disease hemolymphatic stage	(1) Pentamidine	(1) Fexinidazole ¹ (A) Pentamidine (A) Eflornithine ² (R) NECT ³ (R) Melarsoprol	(1) Fexinidazole ¹ ≥6 years, ≥20 kg (1) Suramin <6 years old, <20 kg (A) Pentamidine, acute disease, until suramin can be obtained
	Late-stage disease meningoencephalic stage	(1) Fexinidazole ¹ (nonsevere disease) (1) NECT (severe disease, all ages) (A) Eflornithine ³ , if NECT not available (R) Melarsoprol	(1) Melarsoprol, with oral prednisolone If onset of encephalopathy, discontinue melarsoprol. (A) Fexinidazole if melarsoprol cannot be used
Pregnancy (WHO guidelines) Safety data are very limited Goal: Life-saving therapy Treatment may reduce the likelihood of vertical transmission. Newborns should be examined, undergo blood evaluation for circulating trypanosomes, and be followed clinically.			
1st trimester: Pentamidine, Fexinidazole 2 nd trimester: Nifurtimox, Eflornithine, NECT		Acute, rapidly progressive disease – do not delay treatment.	
¹ Fexinidazole should be administered only if there is a high confidence of long-term (up to 24 months) follow up for early detection of relapse. ² Eflornithine monotherapy ³ NECT=Nifurtimox-Eflornithine Combination Therapy			
(1) First-line, (A) Alternative if preferred drug is not available, (R) Rescue = Treatment in cases of treatment failure with a preferred or alternative agent			

25

Pharmacokinetics and Mechanisms of Drugs for gambiense HAT

	PK Profiles	Proposed Mechanisms of Action
Fexinidazole	Oral with meal, large V_d , high protein binding, $t_{1/2}$ up to 27 h loading dose qd x4d, then maintenance dose qd x6d	Activation by parasite nitroreductase to nitroso free radicals that damage DNA and proteins
Pentamidine	IV, IM Binds plasma and tissue proteins high concentrations in organs poor CNS penetration $t_{1/2}$ 6-10 h 1x daily for 7-10 days	Interference with kinetoplast DNA, inhibits RNA and ribosomal function, and nucleic acid synthesis Kinetoplast: Structure in mitochondria of <i>Trypanosoma</i> and <i>Leishmania</i> essential for survival and energy production
Nifurtimox	Oral , $t_{1/2}$ ~3 h 3x daily x10 days (PK info is limited.)	Like the fexinidazole
Eflornithine	IV every 6h x 7d $t_{1/2}$ 1.5-5 h (PK info is limited.)	- Suicide inhibitor of ornithine decarboxylase - Inhibits polyamine synthesis, cell growth and differentiation

Adverse Effects of Drugs for gambiense HAT

Adverse effects: Common / Potentially Serious (not a complete list)

Fexinidazole	Relatively well tolerated Neuropsychiatric effects (agitation, anxiety, insomnia, hallucination, psychosis) Neutropenia (monitor) QTc prolongation
Pentamidine	Relatively well tolerated Common: nausea, anorexia, dizziness, pruritis, injection site reaction Electrolyte abnormalities / hypoglycemia / renal insufficiency
Nifurtimox	Common (generally reversible) Nausea, vomiting, anorexia, weight loss headache, dizziness, vertigo, insomnia, polyneuropathy
Eflornithine	Frequent, generally reversible: fever, pruritus, hypertension, nausea, vomiting, diarrhea, abdominal pain, headaches, dizziness, hearing loss, seizure, arthralgia, rash, alopecia, and bone marrow suppression – anemia, leukopenia, thrombocytopenia

	Suramin, polyanionic compound	Melarsoprol arsenical
PK	Slow IV infusion, protein binding 99% <i>Does not cross BBB</i> Excreted in urine, detectable up to 3 months t_{1/2} 44-54 days	Slow IV (very irritating) <i>Penetrates CSF</i> Rapid conversion to active trivalent melarsen oxide Terminal t_{1/2} ~35 h
MOA	Inhibits glycolytic enzymes, disrupting energy metabolism Blocks NADH oxidation, impairing respiration and energy production Blocks purinergic signaling, interfering with cellular processes	Melarsen oxide reacts avidly and reversibly with sulfhydryl groups on adjacent carbon atoms, including forming a melarsen oxide-trypanothione complex, which inhibits trypanothione reductase <i>Inactivates many enzymes of both the host and the organism.</i>
Adverse effects	Immediate hypersensitivity reaction with shock nephrotoxicity, peripheral neuropathy, exfoliative dermatitis, bone marrow toxicity Coinfection: <i>Onchocerca volvulus</i> (worm, river blindness) → severe immune reaction <i>Avoid suramin in patients who may have O. volvulus coinfection</i>	Melarsoprol reactive encephalopathy – can be fatal Toxicity to all organs can occur: Heart, liver, kidney, bone marrow, GI tract, nerves (including paralysis) Avoid in pregnancy.

Check your knowledge

What are the therapeutic uses of fexinidazole?

What are the favorable properties of fexinidazole compared to the other anti-HAT drugs?

Why is long-term follow-up after fexinidazole treatment essential?

What are the preferred agents for severe late-stage gambiense HAT and for rhodesiense HAT?

Melarsoprol would be effective in early- and late-stage infection of both HAT diseases so why is it the first-line drug only in late-stage rhodesiense HAT?

Check your knowledge

- What are the mechanisms of action and major toxicities of the drugs for HAT treatment?

Fexinidazole	
Pentamidine	
Nifurtimox	
Eflornithine	
Suramin	
Melarsoprol	

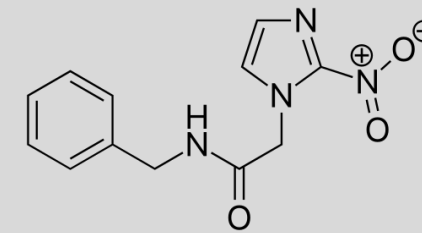
Agents for the treatment of AMERICAN trypanosomiasis

Trypanosoma cruzi

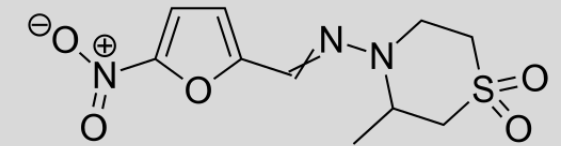
Chagas disease

*Benznidazole

*Nifurtimox



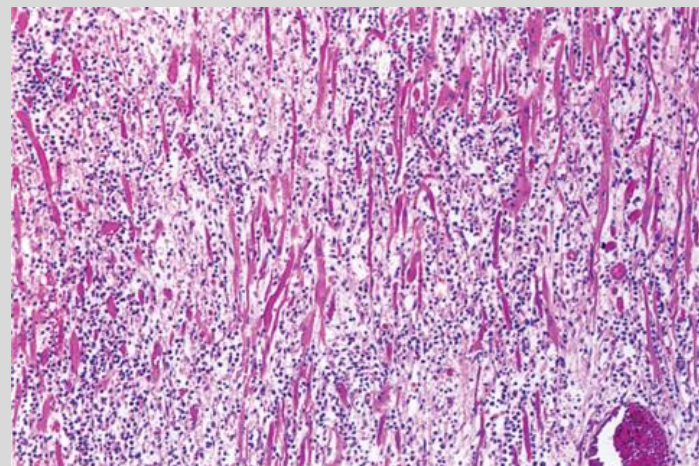
benznidazole



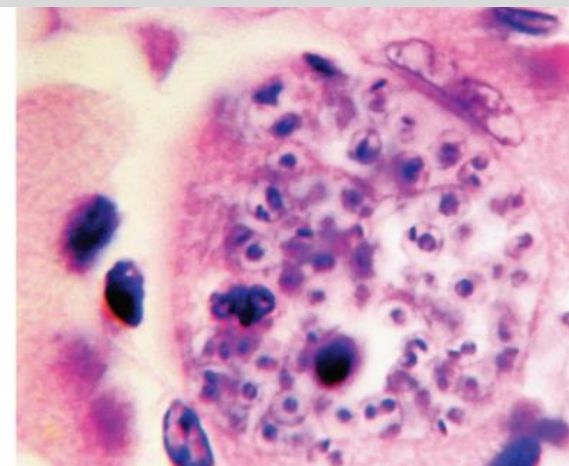
nifurtimox



Source: Kevin J. Knoop, Lawrence B. Stack, Alan B. Storrow, R. Jason Thurman: The Atlas of Emergency Medicine, 5e
Copyright © McGraw Hill. All rights reserved.



A
Source: Kenneth J. Ryan
Sherris & Ryan's Medical Microbiology, 8e
Copyright © McGraw Hill. All rights reserved.



B

Chagas disease.

A. Acute myocarditis with atrophic myofibers separated by inflammatory cells.
B. Trypanosoma cruzi amastigotes clustered in myofiber from the same case.

(Reproduced with permission from Connor DH, Chandler FW, Schwartz DQ, et al: Pathology of Infectious Diseases. Stamford CT: Appleton & Lange; 1997.)

Benznidazole (preferred agent)		Nifurtimox
PK	Oral; eliminated in urine/feces; t ½ ~12h Dosing 2x daily x 60 days	Oral; eliminated in urine as metabolites t½ ~3h Dosing 3x daily x 60 days
Mechanism	Reduced by trypanosomal enzyme trypanosomal nitroreductase → nitro anion free radicals → generate reactive oxygen species that form covalent attachments with parasite’s DNA and other macromolecules → membrane injury, enzyme inactivation, DNA damage	
Effectiveness	Varies with geography, phase, and form of the disease Acute Chagas disease: Cure rates >80% (variable)	
Adverse effects include	Photosensitivity → rash, may progress to severe dermatitis rarely Peripheral neuropathy Bone marrow suppression (rare) Severe effects → interrupt therapy	Common: GI and CNS effects Nausea/vomiting, abdominal discomfort Irritability, disorientation, headache, dizziness, insomnia Less common: paresthesia, polyneuropathy, tremors
Avoid in pregnancy	Animal reproduction studies suggest harm to the fetus. Data on the treatment of acute Chagas disease is limited. There are no specific recommendations for preventing vertical transmission of disease in pregnant patients.	



Source: McPhee SJ, Papadakis MA: *Current Medical Diagnosis and Treatment* 2011, 50th Edition: <http://www.accessmedicine.com>
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Disseminated cutaneous **leishmaniasis** associated with specific anergy induced by *Leishmania aethiopica* infection.

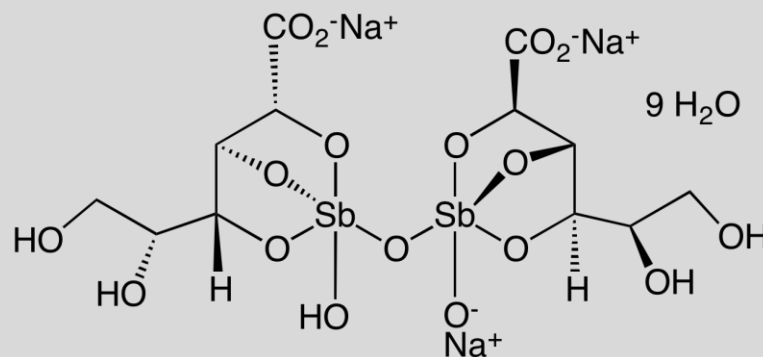


Source: McPhee SJ, Papadakis MA: *Current Medical Diagnosis and Treatment* 2011, 50th Edition: <http://www.accessmedicine.com>
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Chronic nonhealing leishmaniasis recidivans:
Satellite lesions developing around an old sore

Agents for the treatment of leishmaniasis

- * Sodium Stibogluconate / Meglumine Antimoniate
- * Amphotericin B, liposomal
- * Miltefosine
- * Paromomycin



Sodium Stibogluconate

Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	72 Hf
7	87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	104 Rf

Antimony
Sb #51

Leishmaniasis

- Vector borne – sand fly
- *Leishmania* are a heterogenous group of protozoa (Order: Trypanosomatida).
- Endemic in Southern Europe, subtropical, tropical regions
- Obligate intracellular parasites that infect and proliferate in macrophages, monocytes, and other types of phagocytes
- Diseases are characterized by inflammation, infiltration, and necrosis.

Leishmania species produce widely varying clinical syndromes ranging from self-healing cutaneous ulcers to fatal visceral disease.

Visceral leishmaniasis

Mucosal leishmaniasis

Cutaneous leishmaniasis (Old World and New World varieties)

Pharmacology of the Pentavalent Antimonials: Sod. Stibogluconate, Meglumine

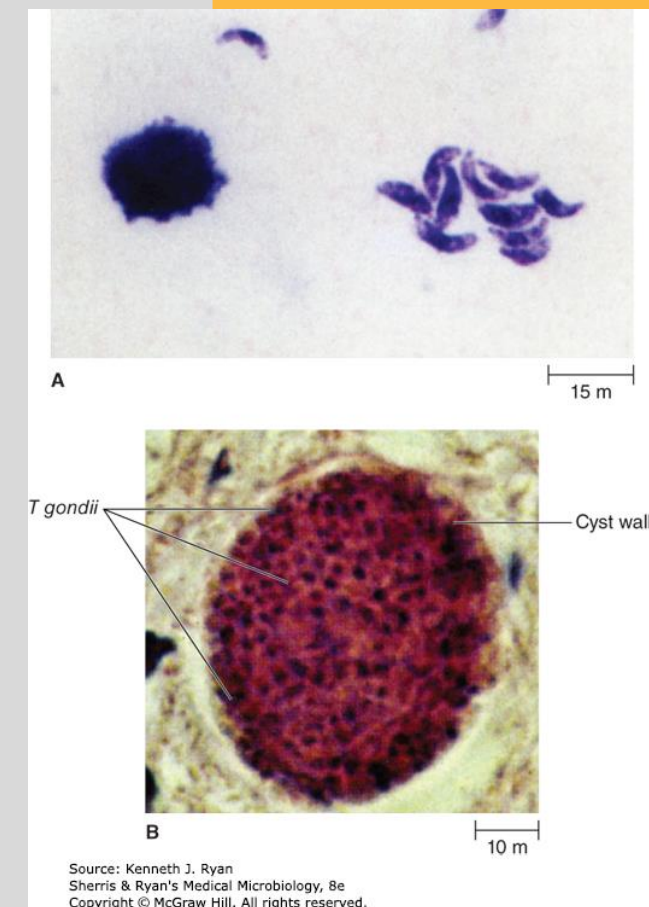
PK	IV, IM sequester in tissues (eg liver, spleen) urine, unchanged $t_{1/2\text{terminal}} \sim 30-80 \text{ h}$
MOA	Mechanism unclear. Likely involves several pathways: Pentavalent antimonial is reduced within the parasite → toxic Sb^{3+} species → may bind thiol groups and interfere with various enzymatic systems → inhibits ATP and GTP availability, disrupts DNA transcription and replication, and other metabolic pathways → kills amastigotes in phagolysosomes of macrophages
Uses	Leishmaniasis: Cutaneous: intralesional injection (also paromomycin topical / nondrug approaches: cryo- and thermotherapy) Mucosal forms: First-line therapy throughout the world Leishmaniasis, visceral (Kala-azar): Alternative to liposomal amphotericin B (AMB is first-line) Efficacy: >95% effective outside certain regions of India
AEs	Relatively well-tolerated. Toxicity increases over the course of therapy (cumulative doses) and is reversible. • Cardiac: nonspecific EKG changes: T wave suppression, QTc interval prolongation • Hematology: Anemia, leukopenia, thrombocytopenia may occur (correct iron deficiency prior to therapy) • Hepatotoxicity; pancreatitis; renal toxicity; muscle aches, nausea, vomiting, fatigue, rash ➤ Practice point: Monitoring, prior to and weekly during therapy: EKG, CBC, and serum creatinine, transaminases, lipase and amylase levels.
Pregnancy	Safe use has not been established. Visceral leishmaniasis is potentially fatal, patients should always be treated without delay.

Drug	Cutaneous leishmaniasis	Visceral leishmaniasis	Post kala-azar dermal lesions
Proposed mechanism			
Amphotericin B liposomal or Deoxycholate (infusion reactions, nephrotoxicity, anemia)	IV daily for 6-7 doses IV 0.5-1 mg/kg daily or every other day for cumulative 15-30 mg/kg dose	- Preferred option, including pregnancy IV to cumulative dose - Secondary prophylaxis (Mediterranean, South Asia, the Americas) IV every 3-4 weeks	IV to cumulative dose
Forms complex with ergosterol in the cell membrane (like the antifungal action), forming pores, causing leakage of cellular contents.			
Miltefosine	Oral TID x 28 days	Oral TID x28 days alternative	Preferred option Oral x12 weeks
Probably interacts with phospholipids and steroids in parasitic cell membranes, inhibition of cytochrome c oxidase which inhibits mitochondrial function, and apoptosis-like death			
Paromomycin protein synthesis inhibitor	Topical ointment or cream	IV or IM x21 days in comb. w miltefosine or antimonial Preferred in East Africa	
Pentamidine Isothionate unknown mechanism	Alternative IV or IM weekly x3 doses Suboptimal efficacy and toxicity	Secondary prophylaxis (east Africa) IV every 3-4 weeks	
Fluconazole Ketoconazole	Oral daily x 6 weeks Oral daily x 28 days		

Agents used in the treatment of toxoplasmosis

*Spiramycin

*Pyrimethamine + leucovorin (folinic acid) + sulfadiazine
(alternatives for sulfadiazine: Atovaquone or Clindamycin)



Toxoplasma gondii. A. Invasive trophozoite forms. B. Cyst in tissue. (Reproduced with permission from Nester EW, Anderson DG, Roberts CE Jr, et al: *Microbiology: A Human Perspective*, 6th ed. New York, NY: McGraw Hill; 2008.)

Toxoplasmosis

Toxoplasma gondii: Cats are natural hosts.

- Toxoplasmosis is an acute, self-limiting illness in most people.
- In pregnancy, vertical transmission to the fetus causes congenital toxoplasmosis.
- Immunocompromised HIV/AIDS patients are at risk for developing toxoplasmic encephalitis.

Pregnancy:

oral Spiramycin (a macrolide antibiotic) at <14 weeks gestation

oral Pyrimethamine + leucovorin + sulfadiazine $\geq$ 14 weeks gestation and documented fetal infection

HIV/AIDS: Treatment of cerebral toxoplasmosis

oral Pyrimethamine + leucovorin + sulfadiazine or IV TMP-SMX

(alternatives if sulfonamide is not tolerated: atovaquone or clindamycin)

Check your knowledge

Which of the following is frequently used for the treatment of leishmaniasis?

- A. Folate synthesis inhibition disrupts DNA synthesis.
- B. Nitroso free radicals generate reactive oxygen species that form covalent attachments with parasite's DNA and other macromolecules causing membrane injury, enzyme inactivation, DNA damage
- C. Protein synthesis inhibition by binding the 16S rRNA of the 30S ribosome, which interferes with the protein synthesis machinery.
- D. Toxic Sb^{3+} interferes with various enzymatic systems, which disrupts ATP and GTP availability, DNA transcription and replication, and other metabolic pathways

What are the drugs in each of the options?

Check your knowledge

Looking at the mechanisms of action of the antiprotozoal drugs, what is the common thread?

What are the preferred drugs for treatment of the following parasitic infections?

- Amoebiasis:
- Gambiense HAT:
- Rhodesiense HAT:
- Chagas disease:
- Mucocutaneous leishmaniasis:
- Visceral leishmaniasis:
- Toxoplasmosis, first trimester:
- Toxoplasmosis, ≥ 14 weeks gestation:

The answers to the Check Your Knowledge questions summarize the pharmacology of the non-plasmodial antiprotozoal drugs.

Check your knowledge

What are the medical justifications for using two different drugs in the treatment of amoebiasis? What are the drugs described in this PowerPoint?

- Metronidazole is a tissue-acting drug to eliminate invasive trophozoites, particularly in cases of amoebic liver abscess or systemic infection.
- Paromomycin (oral) is a luminal agent that acts in the intestines to eliminate amoeba cysts and trophozoites, which prevents recurrence and transmission.

What are their mechanisms of action?

- Metronidazole: Activation by parasite nitroreductase to nitroso free radicals (reactive oxygen species) that damage DNA and proteins
- Paromomycin: Protein synthesis inhibitor, binds 16S rRNA on the 30S ribosome.

What other organisms are susceptible to each of the drugs?

- Metronidazole is active against the protozoa *Trichomonas vaginalis* and *Giardia lamblia* as well *Entamoeba histolytica*.
- FYI: Metronidazole is also effective against obligate anaerobic gram-negative and gram-positive bacteria, including *C. difficile*.
- Paromomycin is active against luminal protozoa *G. lamblia* and *Cryptosporidium* and *Leishmania*, which causes cutaneous, mucosal, and visceral infection.

Check your knowledge

What are the therapeutic uses of fexinidazole?

- All patients ≥ 6 years old, ≥ 20 kg for treatment of early-stage gambiense and rhodesiense HAT and in nonsevere late-stage gambiense HAT.

What are the favorable properties of fexinidazole compared to the other anti-HAT drugs?

- Oral, safer than the other drugs, crosses the blood-brain barrier.

Why is long-term follow-up after fexinidazole treatment essential?

- Follow-up for up to 24 months is required to detect relapse. Treatment failures can occur. Recurrence rates vary based on the patient's adherence to the oral therapy (fexinidazole), severity of the disease, and the patient's immune response.

What are the preferred agents for severe late-stage gambiense HAT and for rhodesiense HAT?

- Gambiense HAT: NECT – nifurtimox eflornithine combination therapy
- Rhodesiense HAT: melarsoprol, the only effective agent

Melarsoprol would be effective in early- and late-stage infection of both HAT diseases so why is it the first-line drug only in late-stage rhodesiense HAT?

- It is very toxic and toxicity occurs frequently. Severe and common effects include heart, liver, kidney, bone marrow, GI tract, nerve toxicities.

Check your knowledge

- What are the mechanisms of action and major toxicities of the drugs for HAT treatment?

Fexinidazole	Activation by parasite nitroreductase to nitroso free radicals (reactive oxygen species) that damage DNA and proteins Neuropsychiatric effects; neutropenia; QTc prolongation
Pentamidine	Possibly interference with kinetoplast DNA, RNA and ribosomal function, and nucleic acid synthesis Electrolyte abnormalities, hypoglycemia, renal insufficiency
Nifurtimox	Activation by parasite nitroreductase to nitroso free radicals that damage DNA and proteins GI complaints, vertigo, insomnia, polyneuropathy
Eflornithine	Suicide inhibitor of ornithine decarboxylase which suppresses polyamine synthesis, cell growth and differentiation Diarrhea, leukopenia, anemia, hair loss, tinnitus, elevated LFTs, dizziness, headache
Suramin	Possibly forms stable complexes with proteins and reacts with a variety of biomolecules Adverse effects are common: immediate hypersensitivity reaction with shock, skin eruptions, vomiting, photophobia, peripheral neuropathy, anaphylaxis, nephrotoxicity
Melarsoprol	Melarsen oxide reacts avidly and reversibly with sulfhydryl groups on adjacent carbon atoms, which inactivates many enzymes of both the host and the organism, including forming a melarsen oxide-trypanothione complex, which inhibits trypanothione reductase Severe effects common: heart, liver, kidney, bone marrow, GI tract, nerve toxicities

Check your knowledge

Which of the following is frequently used for the treatment of leishmaniasis?

- A. Folate synthesis inhibition disrupts DNA synthesis.
 - B. Nitroso free radicals generate reactive oxygen species that form covalent attachments with parasite's DNA and other macromolecules causing membrane injury, enzyme inactivation, DNA damage
 - C. Protein synthesis inhibition by binding the 16S rRNA of the 30S ribosome, which interferes with the protein synthesis machinery.
 - D. Toxic Sb^{3+} interferes with various enzymatic systems, which disrupts ATP and GTP availability, DNA transcription and replication, and other metabolic pathways
- Answer: D. Pentavalent antimonials sodium stibogluconate and meglumine inhibit ATP and GTP availability, disrupt DNA transcription and replication, and other metabolic pathways

What are the drugs in each of the options?

- A. Pyrimethamine: DHFR inhibitor and Sulfadiazine: dihydropteroate synthase (DHPS)
- B. Metronidazole, fexinidazole, nifurtimox, benznidazole
- C. Paromomycin

Check your knowledge

Looking at the mechanisms of action of the antiprotozoal drugs, what is the common thread?

- Antiparasitic drugs generate free radicals and reactive oxygen species during activation, which damage cellular and molecular structures, or complex and inactivate cellular proteins or interfere with metabolism. Thus, parasites die.

What are the preferred drugs for treatment of the following parasitic infections?

- Amoebiasis: metronidazole (systemic); paromomycin (luminal)
- Gambiense HAT: fexinidazole early-stage, nonsevere late-stage ≥ 6 years old; NECT severe late stage
- Rhodesiense HAT: fexinidazole early-stage ≥ 6 years old; melarsoprol late-stage
- Chagas disease: benznidazole (better tolerated); nifurtimox
- Mucocutaneous leishmaniasis: sod. stibogluconate / meglumine
- Visceral leishmaniasis: amphotercin B / pentavalent antimonial / miltefosine (oral)
- Toxoplasmosis, first trimester: spiramycin
- Toxoplasmosis, ≥ 14 weeks gestation: pyrimethamine + sulfadiazine + leucovorin

References

- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics 14e, 2023; Chapter 67: Chemotherapy of Protozoal Infections: Amebiasis, Giardiasis, Trichomoniasis, Trypanosomiasis, Leishmaniasis, and Other Protozoal Infection
- Access Medicine Harrison's Principles of Internal Medicine, 21e, 2023; Chapter 222. Pharmacology of Agents Used to Treat Parasitic Infections and Chapter 227: Chagas Disease and African Trypanosomiasis
- Access Medicine Jawetz, Melnick & Adleberg's Medical Microbiology, 28e, 2019 Chapter 46: Medical Parasitology
- Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 52: Antiprotozoal Drugs
- Access Medicine Sherris & Ryan's Medical Microbiology, 8e, 2022; Chapter 52: Sarcomastigophora: The Amebas; Chapter 53: Sacromastigophora: The Flagellates
- Centers for Disease Control and Prevention <http://www.cdc.gov/parasites/>
- Lexidrugs monographs
- The Sanford Guide to Antimicrobial Therapy, 52e, 2022
- UpToDate: various topics; literature reviews are current
- World Health Organization WHO: Essential Medicines and Health Products Information Portal <https://www.who.int/>
- Feldman R, Jaszczenski R. Can Metronidazole Cause a Disulfiram-Like Reaction? A Case-Control Study Propensity Matched by Age, Sex, and Ethanol Concentration. WMJ. 2023 Jul;122(3):171-177. PMID: 37494646. <https://pubmed.ncbi.nlm.nih.gov/37494646/>

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>